Do CNNs Really Understand Images or Just Textures? A Study Using Image Distortions

https://github.com/kowshikrepalle/24154830

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ABSTRACT

Convolutional Neural Networks (CNNs) are widely used in image classification due to their ability to automatically learn hierarchical representations from raw pixel data. Despite their success, recent research suggests that CNNs may rely more on texture-based patterns than true structural understanding of objects. This study investigates whether CNNs genuinely understand object shapes or depend primarily on texture information. A CNN model was trained on the Fashion-MNIST dataset and evaluated using distorted images, including blurred, noisy, and edge-only representations. The model achieved approximately 86% accuracy on original images, which decreased to around 78% for blurred images, 33% for noisy images, and 42% for edge-only images. The results show that performance drops significantly when texture information is disrupted, particularly under noisy conditions. These findings suggest that CNNs rely heavily on local patterns rather than global shape representations. The study highlights important limitations of CNNs and discusses implications for robustness, real-world applications, and ethical AI.

INTRODUCTION

Machine learning has become an essential tool for analysing complex data and supporting decision-making across domains such as healthcare, finance, and computer vision. Among these models, Convolutional Neural Networks (CNNs) have achieved state-of-the-art performance in image classification. Their ability to automatically learn features from images has made them highly effective and widely used.

CNNs operate by applying convolutional filters to input images, enabling detection of patterns such as edges, textures, and shapes. These features are combined across layers to form increasingly complex representations. As a result, CNNs are often assumed to understand objects meaningfully. However, recent studies suggest that CNNs may rely more on texture rather than shape.

Understanding whether CNNs truly capture object structure is important for evaluating reliability. If CNNs depend heavily on texture, their performance may degrade when images are distorted, even if overall shape remains unchanged. This raises concerns about robustness and generalisation in real-world applications.

This study investigates this issue by evaluating CNN performance under different image distortions. By comparing results across original, blurred, noisy, and edge-only images, it aims to determine whether CNNs rely more on texture or structural features.

BACKGROUND OF CNN

Convolutional Neural Networks are a class of deep learning models specifically designed for image data. A typical CNN architecture consists of convolutional layers, activation functions, pooling layers, and fully connected layers. Convolutional layers apply filters to detect local patterns, while pooling layers reduce spatial dimensions and improve efficiency.

The hierarchical structure of CNNs enables them to learn increasingly complex features. Early layers detect simple features such as edges, while deeper layers capture textures and object parts. This hierarchical feature extraction is a key reason for the success of CNNs (LeCun et al., 2015).

However, CNNs have limitations. Research by Geirhos et al. (2019) shows that CNNs often exhibit a strong texture bias. This means that instead of learning object shape, the model relies on surface patterns. As a result, CNNs may fail when texture information is altered, even if the object structure remains unchanged.

This study builds on this concept by experimentally testing CNN performance under different distortions.

DATASET AND EXPERIMENTAL SETUP

The Fashion-MNIST dataset was used in this study. It consists of 28×28 grayscale images across ten clothing categories. The dataset was chosen because it contains both shape and texture information, making it suitable for analysing CNN behaviour.

A subset of the dataset was used for efficient computation. The CNN model was implemented using PyTorch and consists of two convolutional layers followed by fully connected layers. ReLU activation functions and max-pooling were used for feature extraction.

The model was trained using the Adam optimiser and cross-entropy loss function. Importantly, training was performed only on original images to evaluate how well the model generalises to distorted inputs.

Three distortions were applied to the test data:

- **Gaussian blur** to remove fine texture details
- **Additive noise** to simulate real-world corruption
- **Edge detection** to preserve shape while removing texture

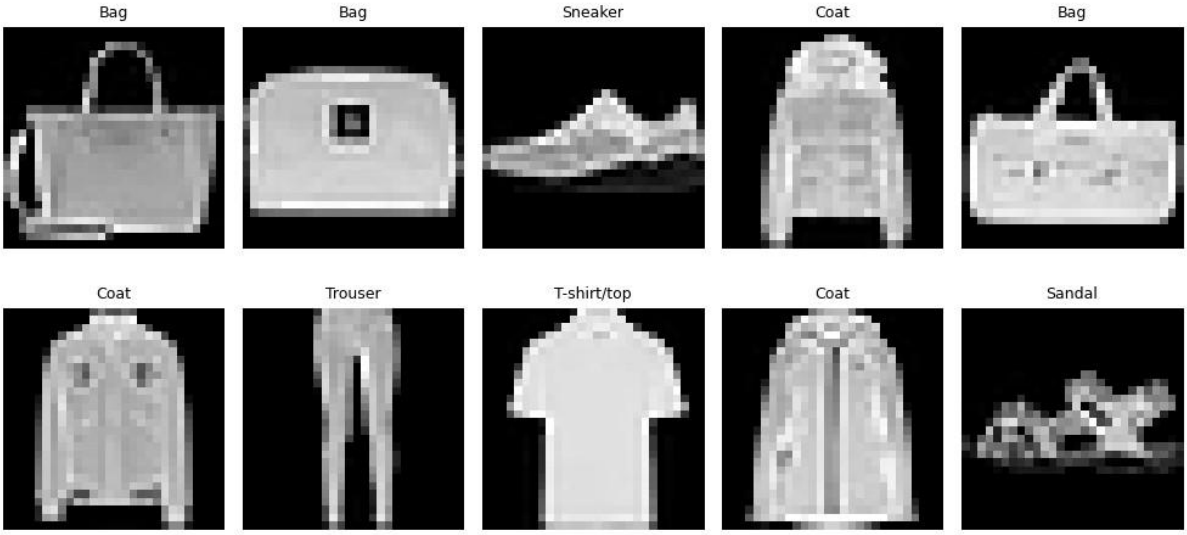
These distortions allow us to isolate different visual features and analyse their impact on classification performance.

RESULTS AND VISUALISATION

1. Dataset Distribution

Figure 1: Sample images from the Fashion-MNIST dataset.

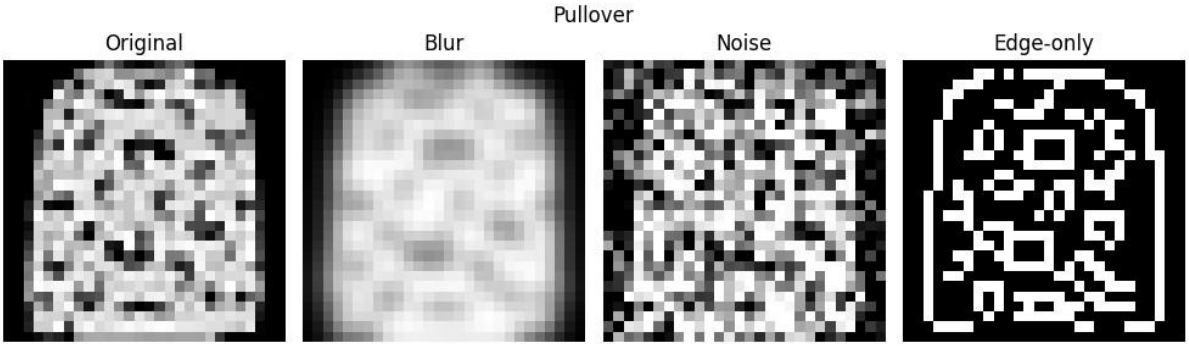
The structure of the dataset is shown in Figure 1, which displays sample images from the FashionMNIST dataset. The images demonstrate variations in both shape and texture across different clothing categories.



2. Image Distortions

Figure 2: Original, blurred, noisy, and edge-only images.

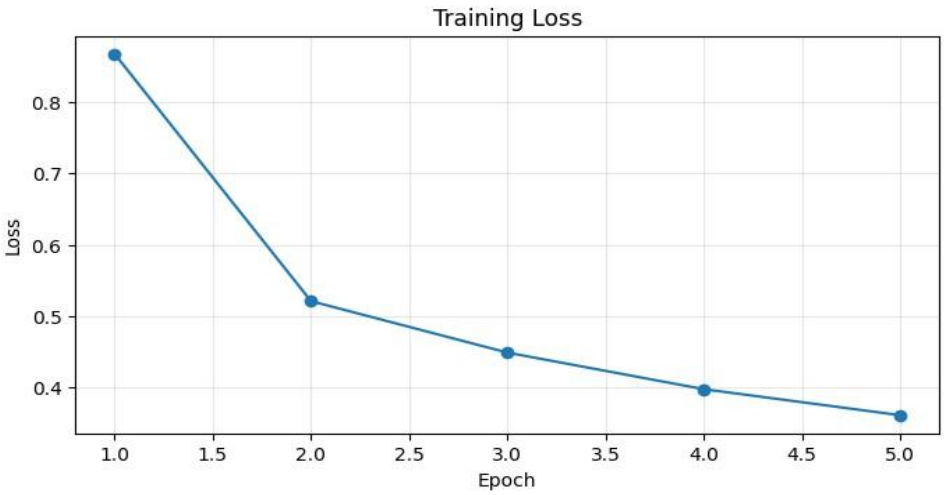
Figure 2 illustrates the distortions applied in this study. Blur removes fine details, noise introduces random variation, and edge detection preserves structural outlines.



3. Training Loss

Figure 3: Training loss over epochs.

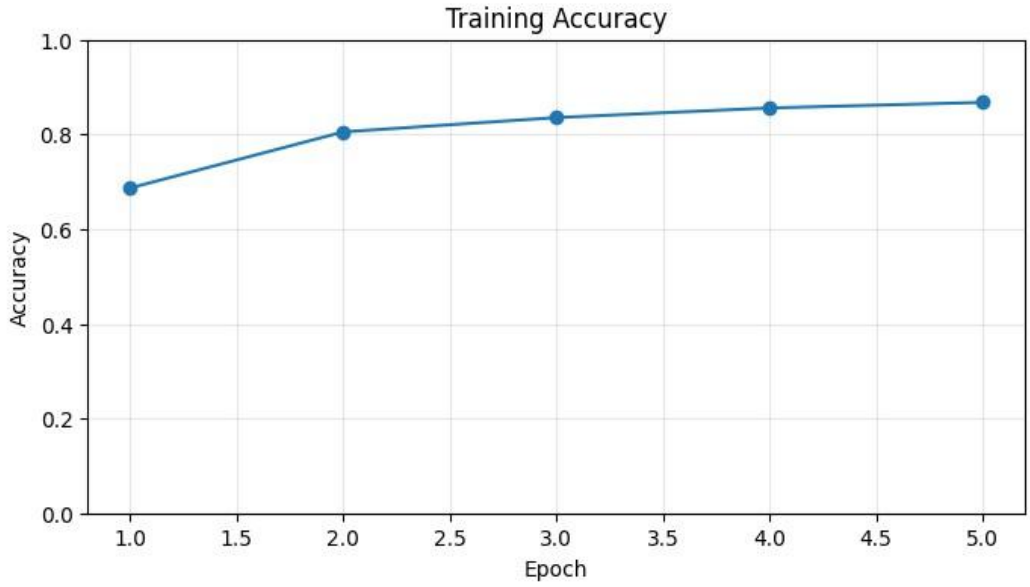
Figure 3 shows the training loss over epochs. The steady decrease indicates successful learning and convergence.



4. Training Accuracy

Figure 4: Training accuracy over epochs

Figure 4 presents training accuracy, which increases over time and confirms effective learning.

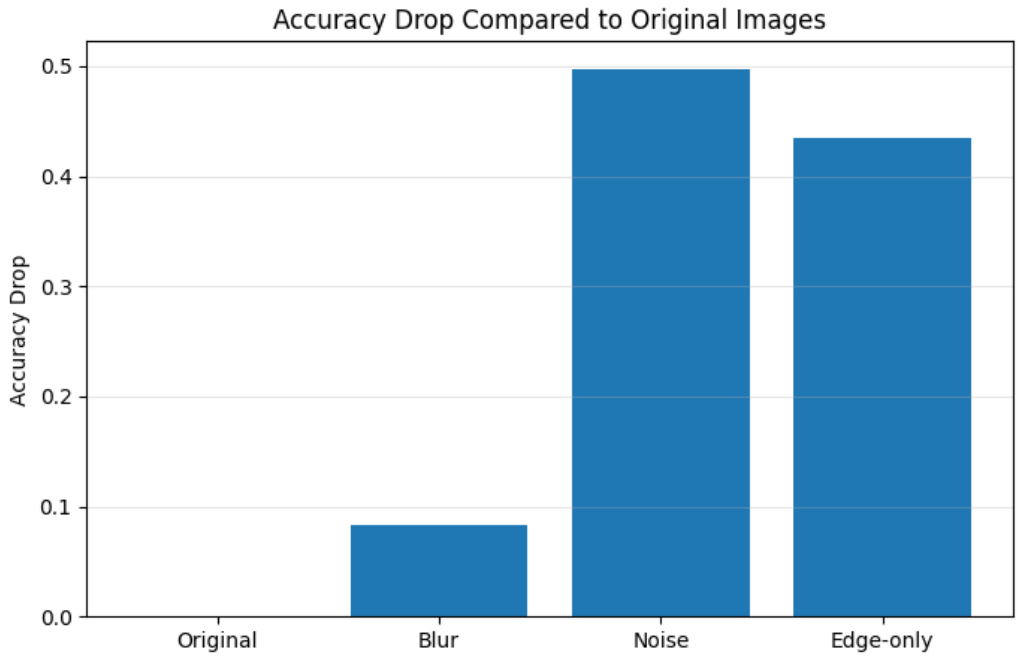


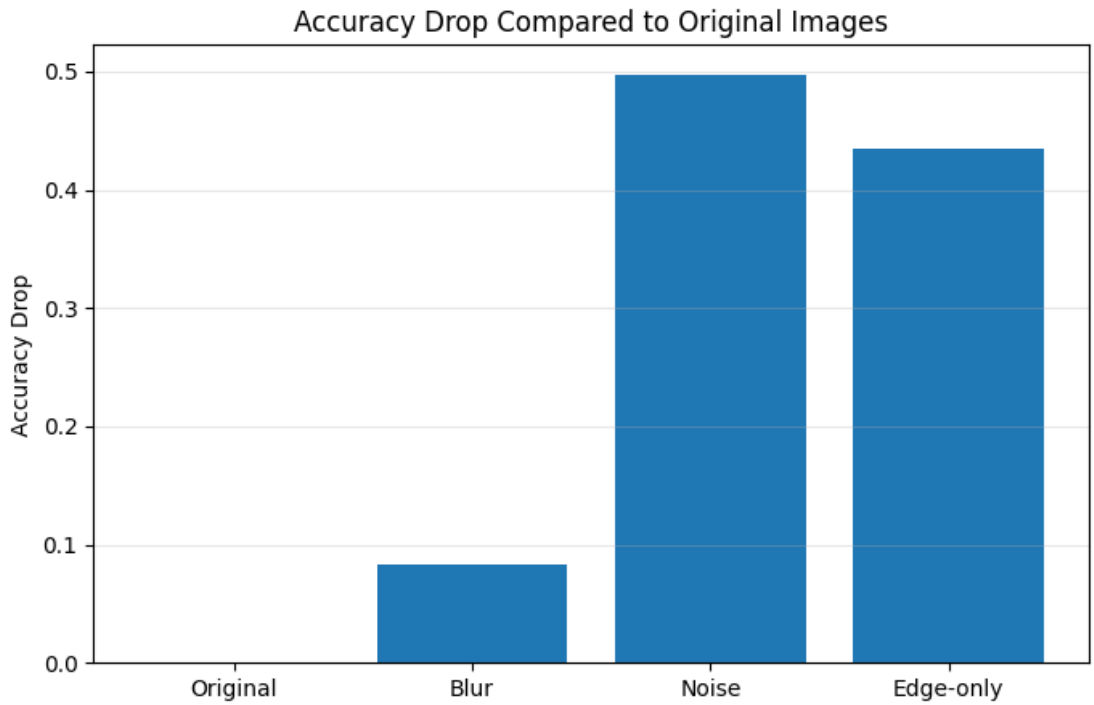
5. Accuracy Comparison

Figure 5 presents a comparison of classification accuracy across different image conditions. The CNN achieves approximately 86% accuracy on original images, demonstrating strong performance when evaluated on data similar to the training set. However, performance decreases under distortion. Accuracy drops to around 78% for blurred images, indicating that removing fine details affects classification moderately.

A much larger drop is observed for noisy images, where accuracy decreases to approximately 33%, representing the most significant degradation in performance. This suggests that the model is highly sensitive to random perturbations in pixel values.

For edge-only images, accuracy is approximately 42%, which is higher than the noisy condition but still significantly lower than the original and blurred cases. Although edge detection preserves the shape of objects, the absence of texture information leads to poor classification performance.





Confusion Matrix Analysis

The classification performance of the CNN is further analysed using confusion matrices, as shown in Figure 6. The confusion matrix provides a breakdown of correct and incorrect predictions across all classes.

For the original images, the confusion matrix shows a strong diagonal pattern, indicating that most predictions are correct. This confirms that the CNN has learned discriminative features from the training data. However, some misclassifications occur between visually similar classes such as shirts, coats, and pullovers.

In contrast, the confusion matrix for distorted images, particularly edge-only inputs, shows a clear increase in off-diagonal values. This indicates that the model struggles to distinguish between classes when texture information is removed. The confusion between similar categories becomes more pronounced, suggesting that the CNN relies heavily on texture-based features.

Interestingly, certain classes such as sandals, sneakers, and bags remain relatively well classified even under distortion, likely due to their distinctive shapes.

Overall, the confusion matrix analysis provides insight into the model's behaviour, demonstrating that errors are systematic rather than random. This further supports the conclusion that CNNs rely more on local texture patterns than global shape understanding.

Figure 6a: Confusion matrix showing classification performance on original images.

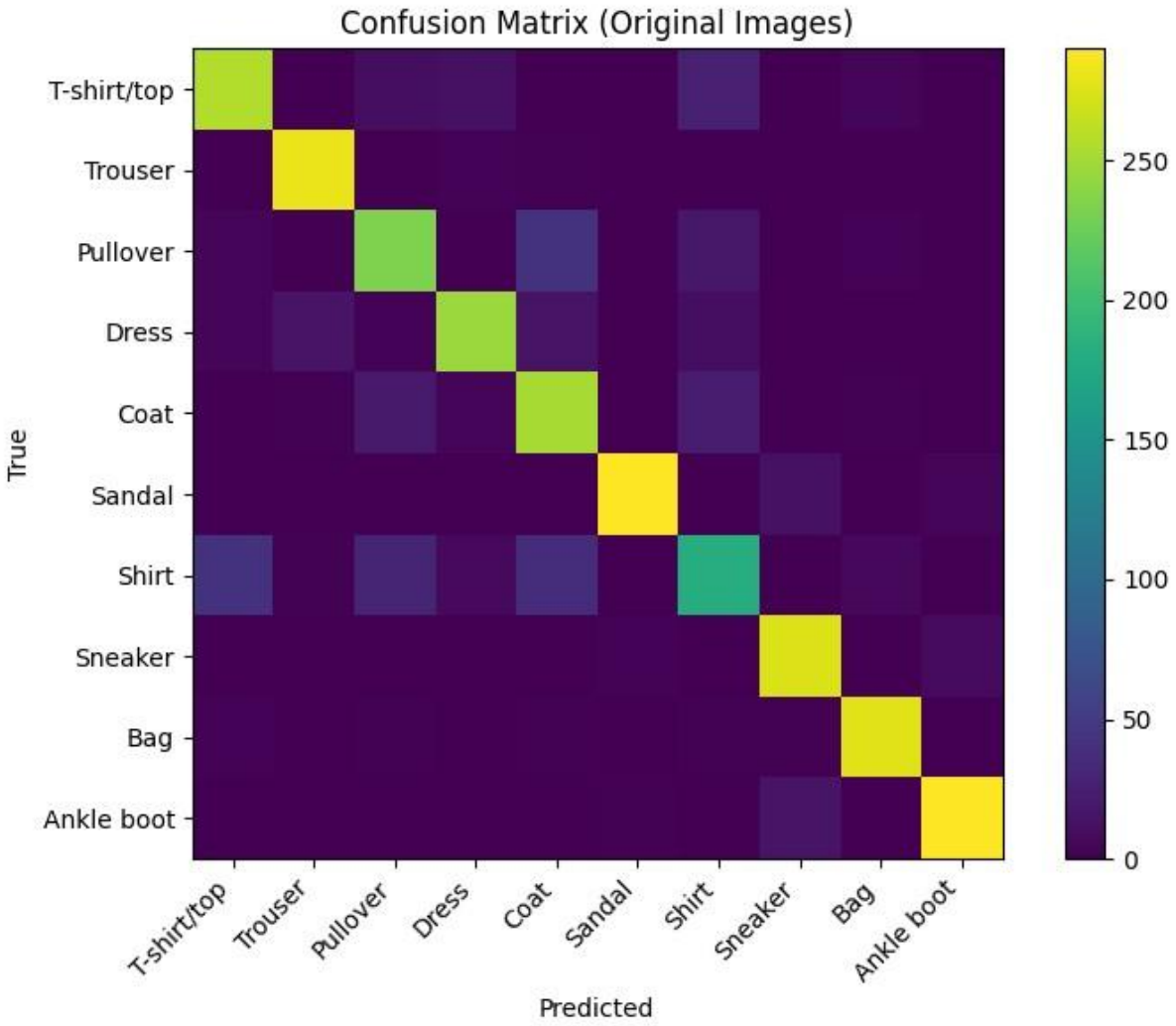
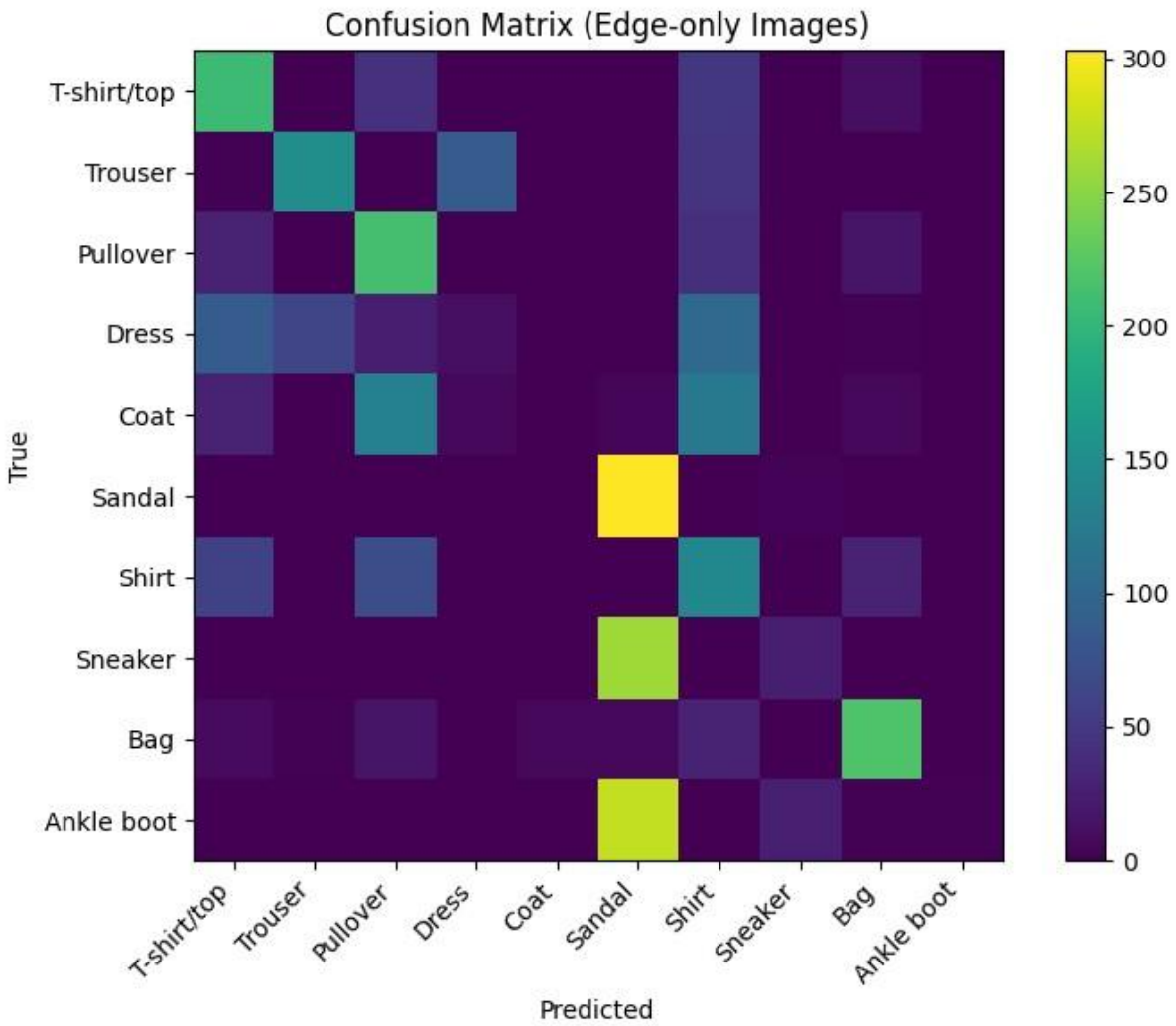


Figure 6b: Confusion matrix showing classification performance on edge-only images.



DISCUSSION

The results demonstrate that CNN performance is significantly affected by image distortions, providing insight into how the model interprets visual information. The model performs best on original images, achieving high accuracy when evaluated on data similar to the training set. However, performance decreases under distortion, indicating limited robustness.

Among the distortions, the most significant drop in performance occurs under noisy conditions, where accuracy falls to approximately 33%. This suggests that the CNN is highly sensitive to random variations in pixel values, which disrupt the learned patterns. Noise introduces unpredictable changes that make it difficult for the model to recognise familiar features.

Performance on blurred images decreases more moderately, indicating that the model can still extract some useful information when fine details are removed. However, the results for edge-only images are particularly important for understanding model behaviour. Despite preserving the overall shape of objects, edge-only images result in low accuracy (approximately 42%), demonstrating that the model struggles when texture information is removed.

This suggests that CNNs do not primarily rely on global shape representations, but instead depend heavily on local texture patterns. This behaviour can be explained by the architecture of CNNs, where convolutional filters operate on small regions and learn statistical patterns rather than abstract object structures.

These findings are consistent with previous research showing that CNNs exhibit a texture bias. While noise produces the largest drop in performance due to its disruptive nature, the edge-only results are particularly important because they isolate shape information. The model's poor performance in this condition provides strong evidence that CNNs rely more on texture than on true structural understanding.

Overall, the results highlight a key limitation of CNNs: their reliance on local patterns reduces their ability to generalise to distorted or unfamiliar inputs. This has important implications for real-world applications where image quality cannot be guaranteed.

LIMITATIONS

This study has several limitations. The Fashion-MNIST dataset is relatively simple and may not represent real-world complexity. The CNN architecture used is shallow, which may limit feature learning. Additionally, the evaluation focuses only on accuracy and does not consider other metrics such as uncertainty or confidence. Distortions were applied only during testing, which may not reflect models trained with data augmentation.

FUTURE WORK

Future work could explore deeper architectures such as ResNet or Vision Transformers. Training with augmented data may improve robustness. Additionally, shape-based learning approaches could help reduce texture bias.

ETHICAL AND REAL-WORLD IMPLICATIONS

The findings highlight important ethical concerns. CNNs that rely on texture may produce incorrect predictions in real-world scenarios where images are distorted. This is particularly critical in applications such as medical diagnosis and autonomous driving. Ensuring robustness and reliability is essential for safe AI deployment.

ACCESSIBILITY CONSIDERATIONS

The report uses grayscale and high-contrast visuals. All figures include clear labels and descriptions, ensuring accessibility.

CONCLUSION

This study investigated whether Convolutional Neural Networks (CNNs) truly understand images or rely primarily on texture-based patterns. The results show that CNN performance decreases significantly under image distortions, with the largest degradation observed under noisy conditions. This indicates that the model is highly sensitive to random variations in pixel values.

Although the edge-only condition does not produce the lowest accuracy overall, it remains particularly important for analysis because it preserves object shape while removing texture information. The model's relatively poor performance in this condition demonstrates that it struggles to classify images when texture cues are absent, even when structural information is still available.

These findings suggest that CNNs rely heavily on local texture patterns rather than global shape representations. While CNNs are highly effective under standard conditions, their reliance on texture limits their ability to generalise to distorted or unfamiliar inputs. This highlights the need for more robust models that can better capture high-level structural features and perform reliably in real-world scenarios.

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